GATE DA 2027

Artificial Intelligence

Topic-wise Question Bank

Weighted by what GATE DA 2024–2026 actually asked

84 practice questions + **6** real previous-year questions • **8** topics

MCQ • MSQ • NAT — the three GATE question types

Answer key at the end — answers only, no worked solutions

Every numerical answer in this book (B^+-tree fan-out, FD closures, SQL counts, z-scores) was recomputed independently in Python before the key was fixed — all 16 machine checks agreed. The 6 previous-year questions carry the official GATE answer key. Practice options are shuffled, so the key is not all "A".

Prepared for Ashutosh • August 2026

How this book is built

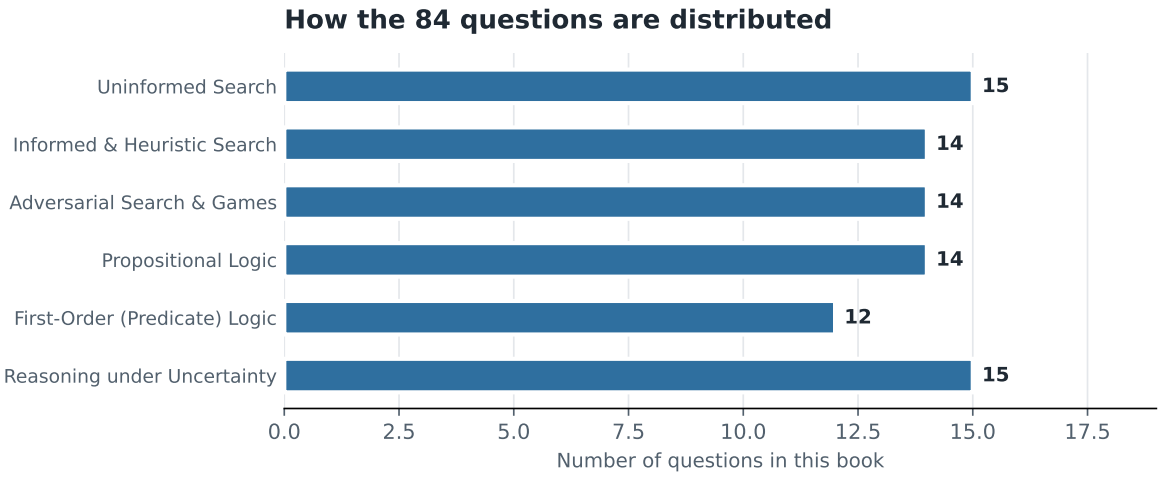
AI is a small but wildly volatile subject — which is exactly why it rewards preparation. Across the official papers its weight swung from about **11 marks in 2024** down to roughly **3 in 2025** and back to about **8 in 2026**. You cannot predict how many marks it will carry in 2027, so the sensible plan is to cover the whole (small) syllabus well: search, logic, and reasoning under uncertainty. The questions are concept-checks and small hand-computations, not proofs.

What GATE DA actually asks in AI

Reading every AI question in the official 2024 and 2026 papers, four habits stand out:

1. **Classify a search algorithm.** "Which is NOT uninformed?" (2026 Q.13, answer A*); "which combination of admissible heuristics is always admissible?" (2024 Q.23); BFS vs DFS behaviour on a small state space (2024 Q.44). Know the properties table cold.
2. **Reason about logic.** What entailment means (2026 Q.24); which quantified first-order sentences are valid (2026 Q.48). These are pure definition checks — easy marks if you know $\forall/\exists$ and entailment.
3. **Compute one probability from a Bayesian network.** Multiply the conditional-probability entries along the network (2024 Q.64, answer 0.125). Shares its machinery with Probability and Machine Learning.
4. **Evaluate a small game tree.** Minimax with MAX/MIN, and when alpha–beta prunes (2026 Q.30). You propagate values up by hand.

The GATE DA AI syllabus is genuinely short — *search* (informed, uninformed, adversarial), *logic* (propositional and first-order), and *reasoning under uncertainty* (Bayesian networks, conditional independence, exact inference by variable elimination, approximate inference by sampling). This book gives each of those six areas its own topic.



Uninformed vs informed search (know this table cold)

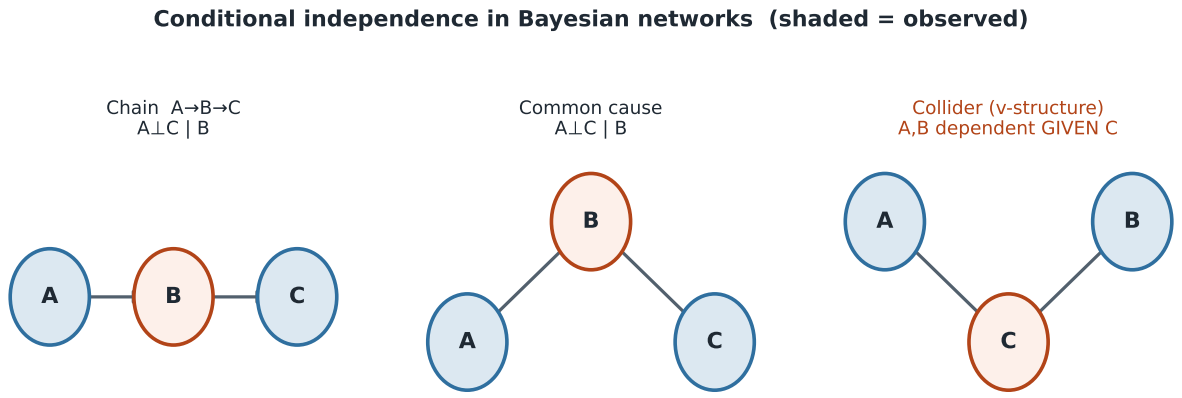
Strategy	Complete?	Optimal?	Time	Uses h?
BFS	Yes	Yes (equal costs)	O(b^d)	no
Uniform-Cost	Yes	Yes	O(b^d)	no
DFS	No (infinite)	No	O(b^m)	no
Iterative Deepening	Yes	Yes (equal costs)	O(b^d)	no
Greedy best-first	No	No	O(b^m)	yes
A*	Yes	Yes (admissible h)	exp.	yes

Question types and marking (same as the real paper)

Type	Description	Negative marking
MCQ	four options, exactly one correct	$-1/3$ (1-mark) or $-2/3$ (2-mark)
MSQ	four options, one *or more* correct; no partial credit	none
NAT	numerical answer, entered on a virtual keypad	none

This bank is 83% MCQ, 7% MSQ and 10% NAT.

Conditional independence — the one picture to memorize



Eighteen facts that answer most AI questions.

- Uninformed search uses no heuristic: BFS, DFS, UCS, IDDFS, depth-limited. A* and greedy are *informed*.
- BFS: complete, optimal for equal step costs, $O(b^d)$ time *and* memory.
- DFS: $O(bm)$ memory, not complete on infinite/cyclic spaces, not optimal.
- Uniform-Cost Search expands lowest $g(n)$; complete and optimal for non-negative costs.
- IDDFS: DFS memory with BFS completeness/optimality.
- A* expands lowest $f = g + h$; optimal with an admissible ($h \leq h^*$) heuristic (consistent for graph search).
- $\max(h_1, h_2)$ of admissible heuristics is admissible and dominates; a dominating heuristic expands fewer nodes.
- $h \equiv 0$ makes A* into UCS; greedy uses h only and is neither complete nor optimal.
- Minimax: MAX takes the max child, MIN the min; assumes an optimal opponent; $O(b^d)$.
- Alpha–beta returns the same value as minimax while pruning; best case $O(b^{d/2})$, depends on move ordering.
- $X \models Y$ iff $X \Rightarrow Y$ is valid iff $X \wedge \neg Y$ is unsatisfiable.
- Valid = true in every model; satisfiable = true in some; unsatisfiable = true in none.
- $P \Rightarrow Q \equiv \neg P \vee Q$; contrapositive $\neg Q \Rightarrow \neg P$; De Morgan $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.
- $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$; $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$.
- $\forall x\, P(x) \Rightarrow \exists x\, P(x)$ is valid; the converse is not.
- A Bayes net factorizes the joint as $\prod_i P(X_i \mid \text{parents}(X_i))$.
- Chain and common-cause give conditional independence given the middle; a collider makes causes dependent once the effect is observed ("explaining away").
- Exact inference = variable elimination (sum out); approximate = rejection sampling, likelihood weighting, Gibbs/MCMC.

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Part A — Practice questions by topic

1. Uninformed Search

15 questions • 13 MCQ, 1 MSQ, 1 NAT • Suggested time: 27 minutes

Q.1 [MCQ • 1 Mark • Easy]

Which of the following is **NOT** an example of uninformed (blind) search? *(GATE DA 2026, Q.13 pattern.)*

(A) Depth-Limited Search
(B) Depth-First Search
(C) A* Search
(D) Breadth-First Search

Q.2 [MCQ • 1 Mark • Moderate]

Breadth-First Search (BFS) on a tree with branching factor b and shallowest goal at depth d has time complexity

(A) $O(d)$ (B) $O(b^d)$ (C) $O(b)$ (D) $O(bd)$

Q.3 [MCQ • 1 Mark • Moderate]

BFS is guaranteed to find the shallowest goal and is *optimal* when

(A) the heuristic is admissible
(B) there is a single goal
(C) the graph is a tree
(D) all step costs are equal

Q.4 [MCQ • 1 Mark • Moderate]

Depth-First Search (DFS) has space complexity

(A) $O(bd)$ (linear in depth)
(B) $O(d^b)$
(C) $O(b^d)$
(D) $O(1)$

Q.5 [MCQ • 1 Mark • Moderate]

Which uninformed strategy is *complete* and *optimal* for non-negative step costs, expanding the node with the lowest path cost $g(n)$?

(A) Depth-First Search
(B) Depth-Limited Search
(C) Greedy Search
(D) Uniform-Cost Search

Q.6 [MCQ • 1 Mark • Moderate]

Iterative Deepening DFS (IDDFS) combines

(A) minimax with pruning
(B) greedy search with backtracking
(C) the speed of A* with a heuristic
(D) the space efficiency of DFS with the completeness/optimality of BFS

Q.7 [MCQ • 2 Marks • Hard]

In a state space where the start state is 1 and each state n has successors $n{+}1$ and $n{+}2$ (expanded in ascending order, no repeats), reaching goal state 6: which is true? *(GATE DA 2024, Q.44 pattern.)*

(A) DFS expands more states than BFS

(B) Both BFS and DFS expand equal numbers of states

(C) Neither BFS nor DFS reaches state 6

(D) BFS expands more states than DFS

Q.8 [MCQ • 1 Mark • Easy]

A search algorithm is *complete* if it

(A) is guaranteed to find a solution when one exists

(B) always finds the optimal solution

(C) runs in polynomial time

(D) uses a heuristic

Q.9 [NAT • 2 Marks • Moderate]

For branching factor $b = 2$ and goal depth $d = 3$, BFS may expand up to $2^0 + 2^1 + 2^2 + 2^3 =$ _____ nodes in the worst case.

Q.10 [MCQ • 1 Mark • Moderate]

DFS is *not* complete in

(A) a tree of depth 1

(B) infinite-depth (or cyclic, without checking) state spaces

(C) any graph

(D) any finite tree

Q.11 [MCQ • 1 Mark • Easy]

Depth-Limited Search (DLS) is DFS with

(A) a heuristic function

(B) a priority queue

(C) a predetermined depth limit ℓ

(D) alpha-beta pruning

Q.12 [MCQ • 1 Mark • Moderate]

Uniform-Cost Search expands nodes in order of

(A) increasing heuristic $h(n)$

(B) decreasing $g(n)$

(C) insertion order

(D) increasing path cost $g(n)$

Q.13 [MCQ • 1 Mark • Moderate]

BFS uses a ___ for the frontier, while DFS uses a ___.

(A) stack; priority queue

(B) queue (FIFO); stack (LIFO)

(C) priority queue; queue

(D) stack; queue

Q.14 [MSQ • 2 Marks • Moderate]

Which uninformed searches are guaranteed *complete* on a finite state space with repeated-state checking?

(A) Breadth-First Search

(B) Uniform-Cost Search

(C) Iterative Deepening DFS

(D) Greedy Best-First Search

Q.15 [MCQ • 1 Mark • Moderate]

The main drawback of BFS compared with DFS is its

(A) exponential memory requirement $O(b^d)$

(B) non-optimality

(C) incompleteness

(D) use of heuristics

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2. Informed & Heuristic Search

14 questions • 12 MCQ, 1 MSQ, 1 NAT • Suggested time: 24 minutes

Q.1 [MCQ • 1 Mark • Easy]

A* search expands the node minimizing

(A) $f(n) = g(n) + h(n)$ (B) $g(n) - h(n)$ (C) $g(n)$ only (D) $h(n)$ only

Q.2 [MCQ • 1 Mark • Moderate]

A heuristic h is *admissible* if, for every node n,

(A) $h(n) = g(n)$

(B) $h(n) \le h^*(n)$ (never overestimates the true cost)

(C) $h(n) = 0$

(D) $h(n) \ge h^*(n)$

Q.3 [MCQ • 1 Mark • Moderate]

A* with an admissible heuristic (tree search) is

(A) incomplete (B) never optimal (C) optimal (D) the same as DFS

Q.4 [MCQ • 2 Marks • Hard]

Let h_1 and h_2 be admissible heuristics used in A*. Which expression is *always* admissible? *(GATE DA 2024, Q.23 pattern.)*

(A) $h_1 + h_2$

(B) h_1/h_2 (with $h_2 \neq 0$)

(C) $|h_1 - h_2|$

(D) $h_1 \times h_2$

Q.5 [MCQ • 1 Mark • Moderate]

A heuristic h is *consistent* (monotone) if for every edge $n \to n'$ of cost c,

(A) $h(n') = 0$ (B) $h(n) = c$ (C) $h(n) \ge c + h(n')$ (D) $h(n) \le c + h(n')$

Q.6 [MCQ • 1 Mark • Moderate]

Every consistent heuristic is

(A) zero (B) admissible (C) inadmissible (D) equal to g

Q.7 [MCQ • 1 Mark • Moderate]

Greedy best-first search expands the node minimizing

(A) $g(n) + h(n)$ (B) $h(n)$ only (C) path length (D) $g(n)$ only

Q.8 [MCQ • 1 Mark • Moderate]

Greedy best-first search is generally

(A) the same as UCS

(B) always optimal

(C) always complete

(D) not optimal and not complete (may get stuck)

Q.9 [MCQ • 1 Mark • Moderate]

If the heuristic $h(n) = 0$ for all n, then A* reduces to

(A) Uniform-Cost Search

(B) Greedy search

(C) Depth-First Search

(D) random search

Q.10 [NAT • 1 Mark • Easy]

Given $g(n) = 4$ and an admissible $h(n) = 3$, the value $f(n)$ used by A* is _____.

Q.11 [MCQ • 1 Mark • Moderate]

The maximum of two admissible heuristics, $\max(h_1, h_2)$, is

(A) equal to $h_1 + h_2$

(B) inadmissible

(C) admissible and dominates both (a better heuristic)

(D) always zero

Q.12 [MCQ • 1 Mark • Moderate]

A heuristic h_2 *dominates* h_1 (both admissible) if $h_2(n) \geq h_1(n)$ for all n. The dominating heuristic generally

(A) expands fewer nodes in A*

(B) is slower per node only

(C) is inadmissible

(D) expands more nodes

Q.13 [MCQ • 1 Mark • Moderate]

For the 8-puzzle, the number of misplaced tiles and the total Manhattan distance are both

(A) admissible heuristics (Manhattan dominates)

(B) equal always

(C) inadmissible

(D) consistent only for BFS

Q.14 [MSQ • 2 Marks • Moderate]

Which statements about A* are correct?

(A) It is optimal for any heuristic whatsoever

(B) It reduces to UCS when $h \equiv 0$

(C) With an admissible heuristic it is optimal (tree search)

(D) With a consistent heuristic it is optimal for graph search

3. Adversarial Search & Games

14 questions • 10 MCQ, 1 MSQ, 3 NAT • Suggested time: 28 minutes

Q.1 [MCQ • 1 Mark • Easy]

In the minimax algorithm, the MAX player chooses the child with the ___ value and MIN chooses the child with the ___ value.

(A) maximum; average (B) average; maximum (C) minimum; maximum (D) maximum; minimum

Q.2 [NAT • 2 Marks • Moderate]

A MAX root has three MIN children with values $\{3, 5\}$, $\{2, 9\}$ and $\{0, 7, 4\}$ (each MIN takes the min of its set). The minimax value at the root is _____.

Q.3 [MCQ • 1 Mark • Moderate]

For the tree above, the MAX player's best move leads to the child with minimax value 3. MIN's values are $3, 2, 0$; MAX picks the ___.

(A) minimum, i.e. 0

(B) first child

(C) maximum of these, i.e. 3

(D) average

Q.4 [MCQ • 1 Mark • Moderate]

Alpha–beta pruning returns

(A) a random value

(B) the same value as full minimax, but examines fewer nodes

(C) only the root value estimate

(D) a different (approximate) value

Q.5 [MCQ • 1 Mark • Moderate]

In alpha–beta pruning, α is

(A) the branching factor

(B) the tree depth

(C) the best value for MIN

(D) the best value found so far for MAX along the path

Q.6 [MCQ • 2 Marks • Hard]

With *optimal* move ordering, alpha–beta pruning reduces the effective branching factor so it can search roughly

(A) twice as deep as plain minimax in the same time ($O(b^{d/2})$)

(B) the same depth

(C) half as deep

(D) infinitely deep

Q.7 [MCQ • 2 Marks • Hard]

A cut-off (pruning) occurs at a MIN node when its value becomes

(A) negative

(B) greater than β

(C) less than or equal to α

(D) equal to the root

Q.8 [MCQ • 1 Mark • Easy]

Minimax assumes the opponent plays

(A) optimally (adversarially)

(B) cooperatively

(C) randomly

(D) never

Q.9 [NAT • 2 Marks • Moderate]

For a MIN root with MAX children of values $\{1, 4\}$, $\{6, 2\}$ (each MAX takes the max), the minimax value at the root is _____.

Q.10 [MCQ • 1 Mark • Moderate]

The time complexity of plain minimax on a game tree of branching factor b and depth d is

(A) $O(d)$ (B) $O(b^{d/2})$ (C) $O(bd)$ (D) $O(b^d)$

Q.11 [MCQ • 1 Mark • Moderate]

Alpha–beta pruning's effectiveness depends heavily on

(A) the number of players

(B) the utility magnitudes

(C) the root value

(D) the order in which successors are examined

Q.12 [MCQ • 1 Mark • Moderate]

An *evaluation function* is used in game-playing when

(A) the tree is too deep to search to terminal states (depth-limited)

(B) the game is trivial

(C) there is only one player

(D) pruning is impossible

Q.13 [NAT • 1 Mark • Easy]

A MAX node with children values $2, 7, 5$ takes value _____.

Q.14 [MSQ • 2 Marks • Moderate]

Which statements about alpha–beta pruning are correct?

(A) Its savings depend on move ordering

(B) It never changes the minimax value at the root

(C) In the best case it examines $O(b^{d/2})$ leaves

(D) It can be applied only to trees of depth 2

4. Propositional Logic

14 questions • 12 MCQ, 1 MSQ, 1 NAT • Suggested time: 24 minutes

Q.1 [MSQ • 2 Marks • Hard]

Sentence X *entails* sentence Y ($X \models Y$) if, whenever X is true, Y is also true. Which statement(s) is/are correct if X entails Y? *(GATE DA 2026, Q.24 pattern.)*

(A) "if Y then X"

(B) $X \Rightarrow Y$ is valid

(C) $X \wedge \neg Y$ is unsatisfiable (FALSE for all models)

(D) "if X then Y"

Q.2 [MCQ • 1 Mark • Moderate]

A propositional sentence is *valid* (a tautology) if it is true

(A) only when all atoms are true

(B) under every interpretation

(C) under no interpretation

(D) under some interpretation

Q.3 [MCQ • 1 Mark • Moderate]

A sentence is *satisfiable* if it is true

(A) only if it is a tautology

(B) under at least one interpretation

(C) under all interpretations

(D) under no interpretation

Q.4 [MCQ • 1 Mark • Moderate]

$P \Rightarrow Q$ is logically equivalent to

(A) $\neg P \wedge Q$ (B) $P \vee \neg Q$ (C) $\neg P \vee Q$ (D) $P \wedge Q$

Q.5 [MCQ • 1 Mark • Moderate]

The contrapositive of $P \Rightarrow Q$ is

(A) $\neg Q \Rightarrow \neg P$

(B) $\neg P \Rightarrow \neg Q$

(C) $Q \Rightarrow P$

(D) $P \Rightarrow \neg Q$

Q.6 [MCQ • 1 Mark • Easy]

By De Morgan's law, $\neg(P \wedge Q)$ equals

(A) $\neg P \vee \neg Q$

(B) $P \vee Q$

(C) $\neg P \wedge \neg Q$

(D) $\neg P \Rightarrow Q$

Q.7 [MCQ • 1 Mark • Moderate]

$X \models Y$ holds if and only if the sentence $X \Rightarrow Y$ is

(A) unsatisfiable

(B) satisfiable

(C) false

(D) valid (a tautology)

Q.8 [MCQ • 1 Mark • Moderate]

A literal is

(A) an atomic proposition or its negation

(B) a tautology

(C) a disjunction of clauses

(D) a conjunction of atoms

Q.9 [MCQ • 1 Mark • Moderate]

Conjunctive Normal Form (CNF) is a

(A) disjunction of conjunctions

(B) negation of a clause

(C) conjunction of disjunctions (clauses) of literals

(D) single literal

Q.10 [MCQ • 1 Mark • Moderate]

The resolution inference rule operates on sentences in

(A) any natural-language form

(B) first-order form only

(C) DNF only

(D) CNF (clausal form)

Q.11 [MCQ • 2 Marks • Hard]

To prove $KB \models \alpha$ by resolution refutation, one shows that

(A) $KB \wedge \alpha$ is valid

(B) $KB \wedge \neg\alpha$ is unsatisfiable (derives the empty clause)

(C) α is a literal

(D) KB is in DNF

Q.12 [NAT • 1 Mark • Moderate]

The truth table of a sentence with n distinct propositional symbols has _____ rows for $n = 3$.

Q.13 [MCQ • 1 Mark • Easy]

$P \wedge (\neg P)$ is

(A) satisfiable

(B) valid

(C) a tautology

(D) unsatisfiable (a contradiction)

Q.14 [MCQ • 1 Mark • Easy]

$P \vee \neg P$ is

(A) satisfiable but not valid

(B) a contradiction

(C) unsatisfiable

(D) valid (a tautology)

5. First-Order (Predicate) Logic

12 questions • 10 MCQ, 2 MSQ, 0 NAT • Suggested time: 24 minutes

Q.1 [MSQ • 2 Marks • Hard]

Let $P(x)$ be a predicate. Which statement(s) is/are **NOT** valid in first-order logic? *(GATE DA 2026, Q.48 pattern.)*

(A) $\exists x\, P(x) \Rightarrow \forall x\, P(x)$

(B) $\forall x\, P(x) \Rightarrow \exists x\, \neg P(x)$

(C) $\forall x\, P(x) \Rightarrow \exists x\, P(x)$

(D) $\exists x\, P(x) \Leftrightarrow \forall x\, P(x)$

Q.2 [MCQ • 1 Mark • Moderate]

$\forall x\, P(x) \Rightarrow \exists x\, P(x)$ is (over a non-empty domain)

(A) invalid (B) valid (C) unsatisfiable (D) a contradiction

Q.3 [MCQ • 1 Mark • Moderate]

The negation $\neg\forall x\, P(x)$ is equivalent to

(A) $\exists x\, \neg P(x)$

(B) $\exists x\, P(x)$

(C) $\forall x\, \neg P(x)$

(D) $\neg\exists x\, P(x)$

Q.4 [MCQ • 1 Mark • Moderate]

The negation $\neg\exists x\, P(x)$ is equivalent to

(A) $\forall x\, P(x)$

(B) $\forall x\, \neg P(x)$

(C) $\exists x\, \neg P(x)$

(D) $\neg\forall x\, P(x)$

Q.5 [MCQ • 1 Mark • Moderate]

"Every student passed" is best written as

(A) $\exists x\, \text{Passed}(x)$

(B) $\exists x\, (\text{Student}(x) \Rightarrow \text{Passed}(x))$

(C) $\forall x\, (\text{Student}(x) \Rightarrow \text{Passed}(x))$

(D) $\forall x\, (\text{Student}(x) \wedge \text{Passed}(x))$

Q.6 [MCQ • 1 Mark • Moderate]

"Some student passed" is best written as

(A) $\exists x\, (\text{Student}(x) \wedge \text{Passed}(x))$

(B) $\forall x\, \text{Passed}(x)$

(C) $\exists x\, (\text{Student}(x) \Rightarrow \text{Passed}(x))$

(D) $\forall x\, (\text{Student}(x) \wedge \text{Passed}(x))$

Q.7 [MCQ • 2 Marks • Hard]

In general, $\forall x\, \exists y\, P(x, y)$ and $\exists y\, \forall x\, P(x, y)$ are

(A) both contradictions

(B) always equivalent

(C) not equivalent; the second is stronger and implies the first (D) both invalid

Q.8 [MCQ • 1 Mark • Moderate]

A *ground* term in first-order logic is one that

(A) is a predicate (B) is always true

(C) contains a quantifier (D) contains no variables

Q.9 [MCQ • 1 Mark • Moderate]

Universal instantiation allows inferring, from $\forall x\, P(x)$,

(A) $\neg P(a)$ (B) nothing

(C) $P(a)$ for any constant a (D) $\exists x\, P(x)$ only

Q.10 [MCQ • 1 Mark • Moderate]

A first-order sentence is *valid* if it is true

(A) in every model (interpretation and domain) (B) in some model

(C) for one constant (D) in the empty domain only

Q.11 [MCQ • 2 Marks • Hard]

The Skolemization step in FOL replaces

(A) a predicate with a term

(B) a clause with a literal

(C) a universal with an existential

(D) an existentially quantified variable with a Skolem constant/function

Q.12 [MSQ • 2 Marks • Moderate]

Which of the following are valid first-order equivalences?

(A) $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$

(B) $\exists x\, P(x) \Rightarrow \forall x\, P(x)$ is valid

(C) $\forall x\, P(x) \Rightarrow \exists x\, P(x)$ is valid (non-empty domain)

(D) $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$

6. Reasoning under Uncertainty

15 questions • 13 MCQ, 0 MSQ, 2 NAT • Suggested time: 28 minutes

Q.1 [MCQ • 1 Mark • Moderate]

A Bayesian network is a directed acyclic graph in which each node is

(A) always binary

(B) independent of all other nodes

(C) a deterministic function

(D) conditionally independent of its non-descendants given its parents

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Q.2 [MCQ • 1 Mark • Moderate]

For a Bayesian network over variables $X_1, \ldots, X_n$, the joint distribution factorizes as

(A) $\prod_i P(X_i)$ (B) $\sum_i P(X_i)$

(C) $\prod_i P(X_i \mid \text{parents}(X_i))$ (D) $P(X_1) \cdots P(X_n \mid X_1)$ only

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Q.3 [NAT • 2 Marks • Hard]

Consider a Bayesian network $U \to W$, $U \to V$, and $V, W \to Z$, with $P(U{=}1) = 0.5$, $P(V{=}1 \mid U{=}1) = 0.5$, $P(W{=}1 \mid U{=}1) = 1$, $P(Z{=}1 \mid V{=}1, W{=}1) = 0.5$. Then $P(U{=}1, V{=}1, W{=}1, Z{=}1)$ is _____ (three decimals). *(GATE DA 2024, Q.64 pattern.)*

Q.4 [MCQ • 1 Mark • Moderate]

In a chain $A \to B \to C$, A and C are

(A) always independent (B) conditionally independent given B

(C) identical (D) never independent

Q.5 [MCQ • 1 Mark • Moderate]

In a common-cause structure $A \leftarrow B \to C$ (B is a parent of both), A and C are

(A) conditionally independent given B (B) deterministic

(C) never independent (D) marginally independent

Q.6 [MCQ • 2 Marks • Hard]

In a *v-structure* (collider) $A \to C \leftarrow B$, A and B are

(A) always independent

(B) marginally independent but become dependent once C is observed

(C) always dependent

(D) conditionally independent given C

Q.7 [MCQ • 1 Mark • Moderate]

Exact inference in Bayesian networks by *variable elimination* works by

(A) ignoring the evidence

(B) maximizing the likelihood

(C) summing out non-query, non-evidence variables one at a time

(D) enumerating all samples

Q.8 [MCQ • 1 Mark • Easy]

Bayes' rule states $P(A \mid B) =$

(A) $P(A) + P(B)$

(B) $P(B \mid A)P(A)$

(C) $\dfrac{P(B \mid A)P(A)}{P(B)}$

(D) $\dfrac{P(A)P(B)}{P(A \mid B)}$

Q.9 [MCQ • 1 Mark • Moderate]

Approximate inference by *rejection sampling* works by

(A) summing out variables exactly

(B) enumerating the full joint

(C) drawing samples from the prior and discarding those inconsistent with the evidence

(D) never using the evidence

Q.10 [MCQ • 1 Mark • Moderate]

The main drawback of rejection sampling is that

(A) it is exact

(B) it rejects most samples when the evidence is unlikely

(C) it needs no samples

(D) it ignores the prior

Q.11 [MCQ • 2 Marks • Hard]

Likelihood weighting improves on rejection sampling by

(A) ignoring the evidence

(B) rejecting more samples

(C) using exact inference

(D) fixing the evidence variables and weighting each sample by its likelihood

Q.12 [MCQ • 1 Mark • Moderate]

A Bayesian network with n binary variables, each having at most k parents, needs at most $n \cdot 2^k$ parameters instead of the full joint's

(A) n^2 (B) $2^n - 1$ (C) $n!$ (D) $2n$

Q.13 [NAT • 1 Mark • Easy]

For independent Bernoulli variables with $P(A{=}1) = 0.5$ and $P(B{=}1) = 0.5$, $P(A{=}1, B{=}1)$ is _____ (two decimals).

Q.14 [MCQ • 2 Marks • Hard]

Markov Chain Monte Carlo (e.g. Gibbs sampling) generates samples by

(A) exact summation

(B) rejecting inconsistent samples

(C) resampling each variable in turn from its distribution given the current values of the others

(D) drawing all variables independently

Q.15 [MCQ • 1 Mark • Moderate]

In a Bayesian network, observing a common effect (collider) can make two previously independent causes dependent. This phenomenon is called

(A) smoothing (B) marginalization (C) explaining away (D) normalization

Part B — Previous-year questions (with official keys)

These are **genuine questions from the official GATE DA papers** in your archive. Each carries the **official GATE answer key** and its options are kept in the original paper order (not shuffled). Every question is tagged with its exact year and number. Grouped under the same topics as Part A.

1. Uninformed Search

Q.1 PYQ [MCQ • 1 Mark • Easy]

Which of the following algorithms is NOT an example of uninformed search?

(A) Breadth First Search (B) Depth First Search

(C) A* Search (D) Depth-limited Search

Source: GATE DA 2026, Q.13 (1 mark) — official paper & official answer key

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Q.2 PYQ [MCQ • 2 Marks • Hard]

The start state is numbered 1. The successor function for state n returns states $n+1$ and $n+2$. States in the unexpanded list are expanded in ascending order of number, and previously expanded states are not re-added. Which one statement about BFS and DFS is true when reaching goal state 6?

(A) BFS expands more states than DFS (B) DFS expands more states than BFS

(C) Both BFS and DFS expand an equal number of states (D) Both BFS and DFS do not reach goal state 6

Source: GATE DA 2024, Q.44 (2 marks) — official paper & official answer key

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2. Informed & Heuristic Search

Q.1 PYQ [MCQ • 1 Mark • Hard]

Let h_1 and h_2 be two admissible heuristics used in A* search. Which one of the following expressions is always an admissible heuristic?

(A) $h_1 + h_2$ (B) $h_1 \times h_2$

(C) h_1/h_2 (with $h_2 \neq 0$) (D) $|h_1 - h_2|$

Source: GATE DA 2024, Q.23 (1 mark) — official paper & official answer key

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4. Propositional Logic

Q.1 PYQ [MSQ • 1 Mark • Moderate]

Sentence X is said to entail sentence Y if whenever X is TRUE, Y must also be TRUE. Which of the following statements is/are correct if X entails Y?

(A) $X \Rightarrow Y$ (B) $X \wedge \neg Y$ is FALSE

(C) if X then Y (D) if Y then X

Source: GATE DA 2026, Q.24 (1 mark) — official paper & official answer key

5. First-Order (Predicate) Logic

Q.1 **PYQ** [MSQ • 2 Marks • Hard]

Let $P(x)$ be a predicate. Which of the following statements is/are NOT valid in first-order logic?

(A) $\forall x\, P(x) \Rightarrow \exists x\, P(x)$ (B) $\exists x\, P(x) \Rightarrow \forall x\, P(x)$

(C) $\exists x\, P(x) \Leftrightarrow \forall x\, P(x)$ (D) $\forall x\, P(x) \Rightarrow \exists x\, \neg P(x)$

Source: GATE DA 2026, Q.48 (2 marks) — official paper & official answer key

6. Reasoning under Uncertainty

Q.1 **PYQ** [NAT • 2 Marks • Hard]

A Bayesian network has four Bernoulli variables U, V, W, Z with $P(U{=}1) = 0.5$, $P(V{=}1 \mid U{=}1) = 0.5$, $P(W{=}1 \mid U{=}1) = 1$, and $P(Z{=}1 \mid V{=}1, W{=}1) = 0.5$. The value of $P(U{=}1, V{=}1, W{=}1, Z{=}1)$ is _____ (three decimals).

Source: GATE DA 2024, Q.64 (2 marks) — official paper & official answer key

Answer Key

Answers only. A key with several letters (e.g. **AB**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted.

Part A — Practice

1. Uninformed Search

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	B	**3**	D	**4**	A	**5**	D
6	D	**7**	B	**8**	A	**9**	15	**10**	B
11	C	**12**	D	**13**	B	**14**	ABC	**15**	A

2. Informed & Heuristic Search

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	B	**3**	C	**4**	C	**5**	D
6	B	**7**	B	**8**	D	**9**	A	**10**	7
11	C	**12**	A	**13**	A	**14**	BCD		

3. Adversarial Search & Games

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	3	**3**	C	**4**	B	**5**	D
6	A	**7**	C	**8**	A	**9**	4	**10**	D
11	D	**12**	A	**13**	7	**14**	ABC		

4. Propositional Logic

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	BCD	**2**	B	**3**	B	**4**	C	**5**	A
6	A	**7**	D	**8**	A	**9**	C	**10**	D
11	B	**12**	8	**13**	D	**14**	D		

5. First-Order (Predicate) Logic

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	ABD	**2**	B	**3**	A	**4**	B	**5**	C
6	A	**7**	C	**8**	D	**9**	C	**10**	A
11	D	**12**	ACD						

6. Reasoning under Uncertainty

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	C	**3**	0.125	**4**	B	**5**	A
6	B	**7**	C	**8**	C	**9**	C	**10**	B
11	D	**12**	B	**13**	0.25	**14**	C	**15**	C

Part B — Previous-year questions (official keys)

Source	Q	Official answer
GATE DA 2026, Q.13 (1 mark)	A1.1	**C**
GATE DA 2024, Q.44 (2 marks)	A1.2	**C**
GATE DA 2024, Q.23 (1 mark)	A2.1	**D**
GATE DA 2026, Q.24 (1 mark)	A4.1	**ABC**
GATE DA 2026, Q.48 (2 marks)	A5.1	**BCD**
GATE DA 2024, Q.64 (2 marks)	A6.1	**0.125**

Score sheet

Anything below 70% on a topic means the concept, not the practice, is the problem — go back and revise before re-attempting.

Topic	Questions	Correct	Time taken
Uninformed Search	15		
Informed & Heuristic Search	14		
Adversarial Search & Games	14		
Propositional Logic	14		
First-Order (Predicate) Logic	12		
Reasoning under Uncertainty	15		
Total	**84**		